Pathology: Restrictive Lung Diseases – Sarcoidosis and Miscellaneous Interstitial Lung Disorders

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Session Objectives

1. Explain **sarcoidosis** and evaluate its etiology and immunology as well as its clinical, microscopic (specifically granulomatous nature), and gross features
2. Outline the **markers** of **activities** and **differential** diagnosis of sarcoidosis
3. Illustrate **pulmonary Langerhans cell histiocytosis** in terms of clinicopathological features, morphological pathological features, and electron microscopic findings
4. Identify **hypersensitivity pneumonitis** in terms of etiology, histopathological findings, and clinical correlations
5. Distinguish **pulmonary eosinophilic syndromes** and their main diagnostic criteria as well as **pulmonary alveolar proteinosis**

Restrictive/Infiltrative/Interstitial Lung Disease features

- ↓ **Diffusing capacity**, ↑ **A-a gradient**
- Normal or mildly increased **FEV1/FVC** ratio &
  - ↓**TLC**, FVC, lung compliance

- Example diagnoses:
  - Sarcoidoses
  - IPF
  - Granulomatosis with polyangiitis
  - Goodpasture syndrome
  - Hypersensitivity pneumonitis
  - Langerhan cell histiocytosis
  - ARDS, neonatal respiratory distress syndrome
  - Pneumoconioses (Occupational Lung diseases)
  - Drug toxicity (ex. bleomycin, busulfan, amiodarone, methotrexate)

Sarcoidosis

- Systemic **non-necrotizing granulomatous** disease
- **Lung** involvement and bilateral hilar lymphadenopathy in 90% of cases
- Disease of disordered immune regulation in genetically predisposed people
  - Age range **20 – 40** years, **female**:male = 2:1
  - Incidence: US whites 10.9/100,000; **African-Americans** 35.5/100,000

Sarcoidosis is a multisystem disorder of unknown cause

- Exposure to antigen?
- Genetic predisposition?
- Unknown cause but cellular mechanisms are known with **evidence of disordered immune regulation** (in genetically susceptible people) resulting from an immunological response to antigenic trigger, such as **noncaseating granulomas**

  - Clinical diagnosis shows multiple clinical presentations: **Dx of exclusion**
    - **Infectious and noninfectious causes** of granulomatous disease must be ruled out (e.g., tuberculosis)

    - Diagnosis established when **clinicoradiological** findings are supported by **histological** evidence of noncaseating (non-necrotizing) epithelioid (plump macrophages) cell granulomas

Sarcoidosis immunology suggestive of cell-mediated response to unidentified Ag

- Involvement of **CD4+** T cells
- ↑ **cytokines**: IL-2, IL-8, IFNλ, TNF → ↑T cells & monocyte/macrophage activation
- Polyclonal **hypergammaglobulin**emia
- Impaired **dendritic** cell function

Sarcoidosis immunologic diagnosis

- **Tuberculin skin test** for cell mediated immunity ↓
- Immunophenotyping of peripheral blood or **BALF** (Bronchoalveolar Lavage Fluid) **cells:**
  - ↑ **CD4/CD8** Ratio
  - ↑ **total cell count** & number of **lymphocytes > 16%**
  - Nearly normal percentage of eosinophils and neutrophils and absence of plasma cells, distinguish most likely diagnosis of sarcoidosis from most common interstitial lung diseases
- **Noncaseating granulomas**

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2026, Ch. 15

Sarcoidosis multiple clinical presentations and organ involvement incidence

- **Lungs** ~ 90%, larynx, trachea, bronchi & pleural effusions
- **Lymph nodes** ~ 90% (spleen in 3/4 of cases) - chest, neck, under the chin, armpits, or groin
- Liver - 80% biopsy, 20% palp.
- Skin - 25% (skin manifestation lupus pernio), also assoc. w/ erythema nodosum
- Musculoskeletal - 25-39%
- Blood - 4-20% anemia, 40% leucopenia
- Hyper Ca^{2+} - emia / uria - 10%
- Ocular - 11-83% w/ uveitis, keratoconjunctivitis sicca
- Nervous system - 10%, CNS, cranial nerve 7, hypothalamus, pituitary region
- Heart- 5% w/ arrhythmias
- Parotid glands - 6%
- Reproductive organs - <1%, uterus, breast, testis (1/3 of orchiectomies)
- GI tract ~ 1%, mimics Crohns

Extrapulmonary manifestations documented in 30-50% of sarcoidosis cases overall

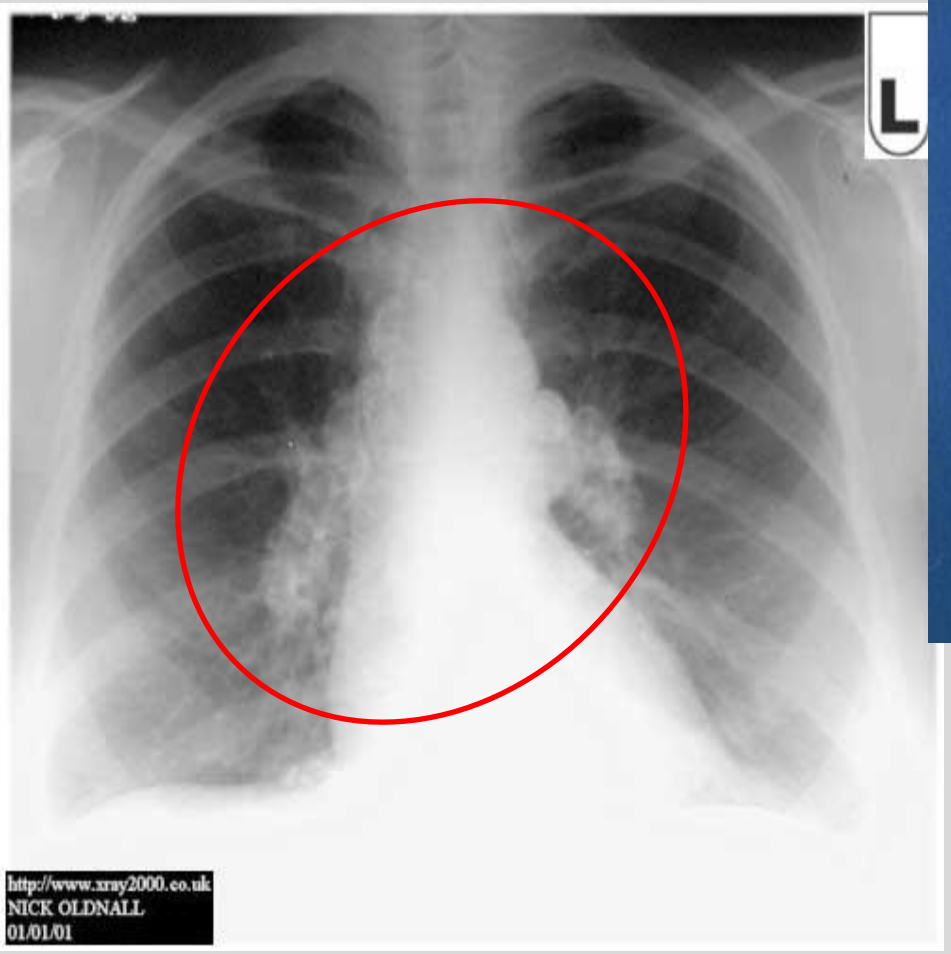
Respiratory Tract

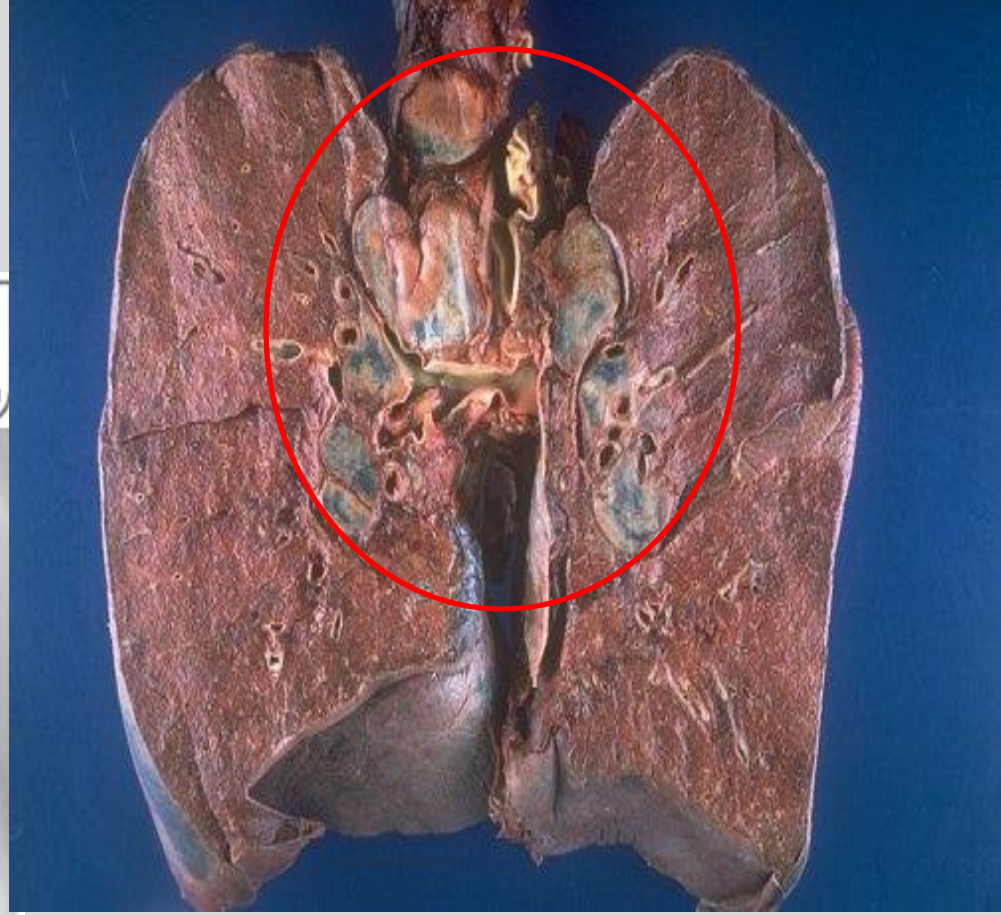
- Occurs at sometime in **essentially all** patients
  - 90% as sole manifestation of sarcoidosis
- **Interstitial** lung disorder involving alveoli (T-cell **alveolitis**), blood vessels, bronchioles
- Transbronchial lung **biopsy diagnostic** in >80% of patients

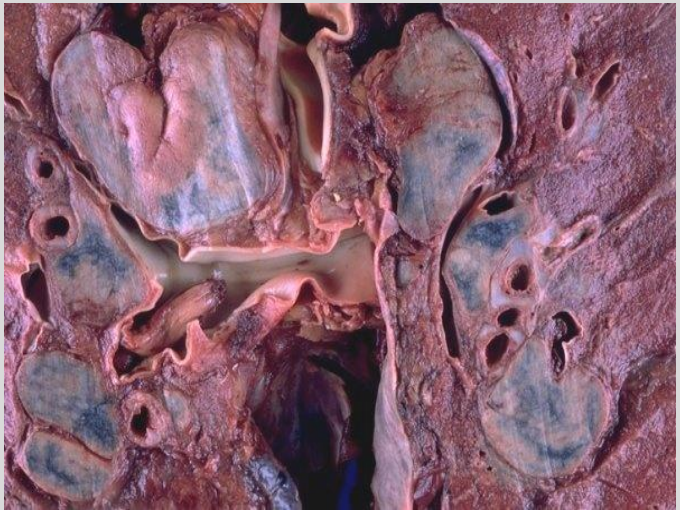
Sarcoidosis

- May be found **incidentally** on X-ray as **bilateral hilar lymphadenopathy**
- Typically present with dyspnea, cough, chest pain, hemoptysis, fever, fatigue, night sweats, weight loss

Bilateral hilar lymphadenopathy







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https://commons.wikimedia.org/wiki/Category:Images_by_Yale_Rosen_MD.

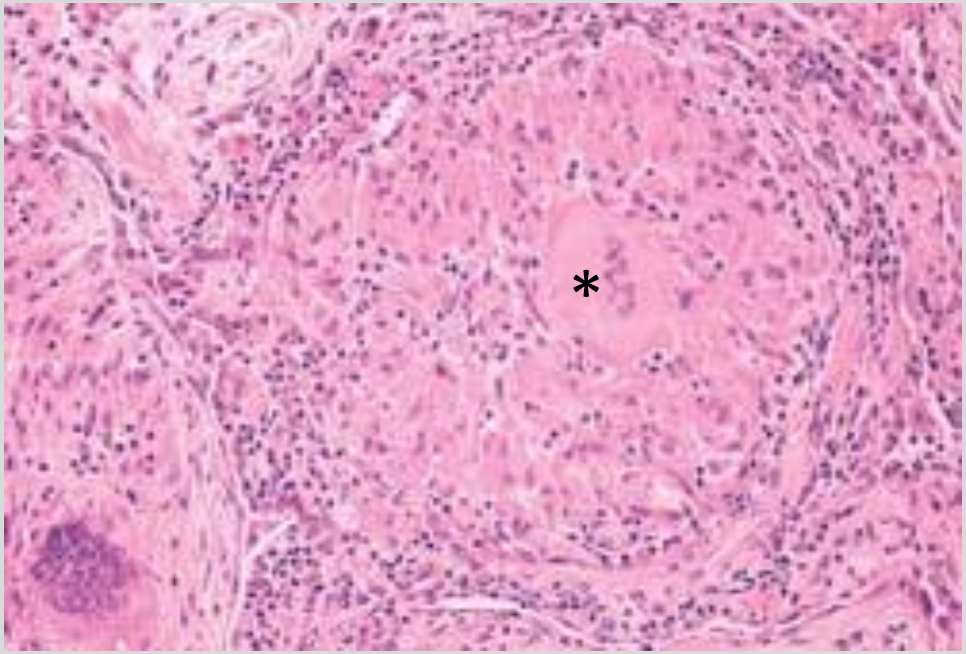
Some differential diagnoses to consider

- **Infections:**
  - Mycobacteria
  - Coccidioidomycosis
  - Histoplasmosis
  - Brucellosis
  - Tularemia
  - Syphilis
  - Toxoplasmosis
- **Malignancy**
  - Lymphoma

- **Autoimmune**
  - Granulomatosis with polyangiitis
  - Churg-Strauss
  - Primary biliary cirrhosis
- **Other**
  - Hypersensitivity pneumonitis
  - Toxic exposures
    - Talc
    - Beryllium
  - Methotrexate toxicity

Diagnosis of exclusion

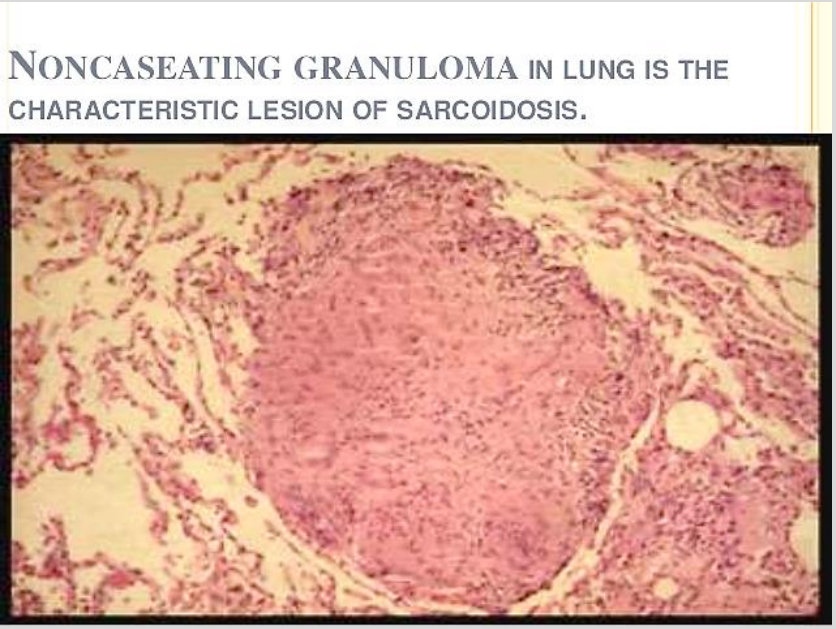
Non-caseating granuloma

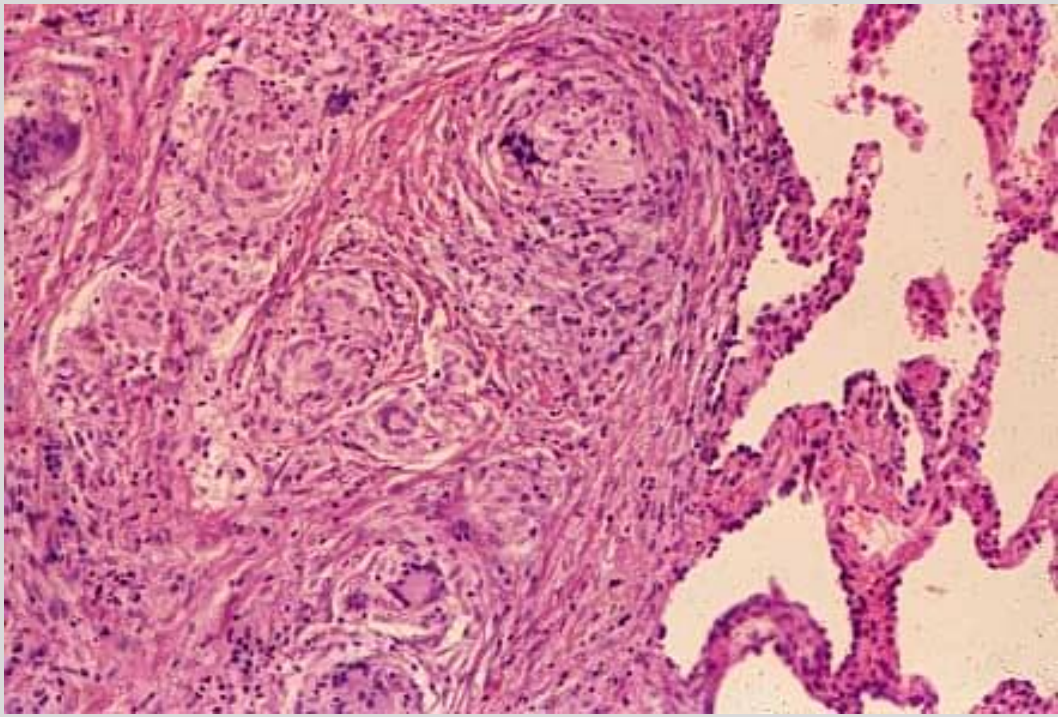


- H&E stained tissue
- **Macrophages**(clustered epithelioid cells & giant cells* & **lymphocytes**
- Chronic: May be replaced by fibrous, hyalinized **scars**

http://medicinembbs.blogspot.com/2011/02/pneumoconiosis.html

Sarcoidosis histopathology – lung with interstitial non-caseating granulomatous inflammation

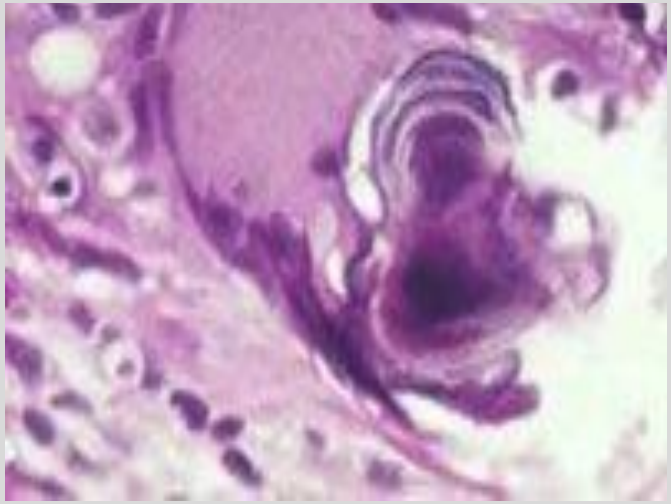




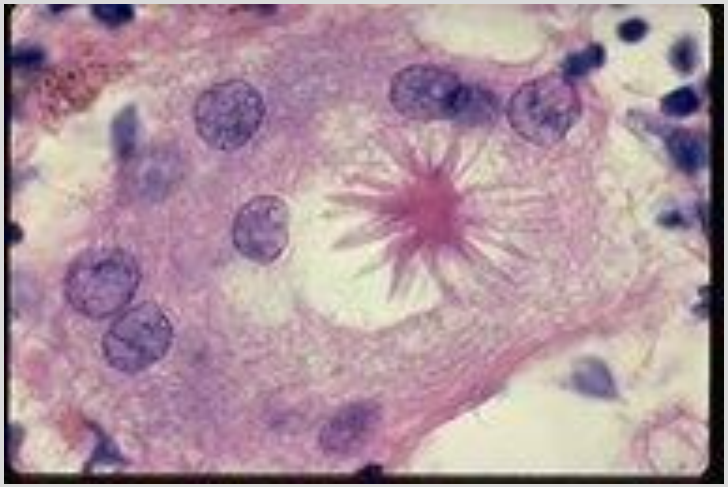
- H&Es: Giant cells and histiocytes (tissue macrophages) form nodular aggregates without necrosis

http://www.slideshare.net/Tareqchowdhury/sarcoidosis-27004011

Sarcoidosis granuloma inclusions are often found, but **not pathognomonic** - found in other granuloma types, H&E stains



- **Schaumann** bodies – basophilic laminated concretions made up of calcium and proteins



- **Asteroid** bodies – stellate inclusions enclosed in giant cells

Sarcoidosis - markers of activity

- Recent progressive **dry cough**
- Progressive **dyspnea**
- **Systemic** signs/symptoms: Fatigue, fever, polyarthralgia, erythema nodosum, lymphadenopathy
- Progressive changes on chest **X-rays** or lung CT scans
- **BALF** - CD4+ alveolitis
- Hypercalciuria > hypercalcemia
- **High** serum levels of **ACE**

Sarcoidosis markers of activity - ACE (angiotensin-converting enzyme)

- **ACE activity** ↑in sarcoidosis**: High serum** levels in ~ 50-80% of patients
  - Support a diagnosis of sarcoidosis
  - In sarcoidosis, produced by granuloma epithelioid cells & macrophages
- Normal level does not exclude sarcoidosis - ↑ ACE also seen in **other diseases**
- Serum ACE seems to reflects **activity** of disease - ↓ in some patients receiving prednisone

Sarcoidosis clinical course

- Unpredictable
  - May be **progressive**, or periods of activity with **remissions**, remission may be permanent and may be **spontaneous**, or induced by **steroid** therapy

- **65-70% recover** with minimal or no residual manifestations
- 20% have permanent loss of some lung function or visual impairment
- 10 –15% die of cardiac or CNS damage, progressive pulmonary fibrosis and cor pulmonale

Pulmonary Langerhans Cell Histiocytosis (LCH)

- **Rare** disease that involves **clonal proliferation** & collections of Langerhans cells, a type of immune cell in skin & mucosal tissues

- Langerhans cells – immature **dendritic** cells with **grooved**, indented nuclei and abundant cytoplasm and an **immunophenotype** of +S100, +CD1a, +CD207 (langerin), -CD68 (monocytes, macrophages)
- Eosinophils often present

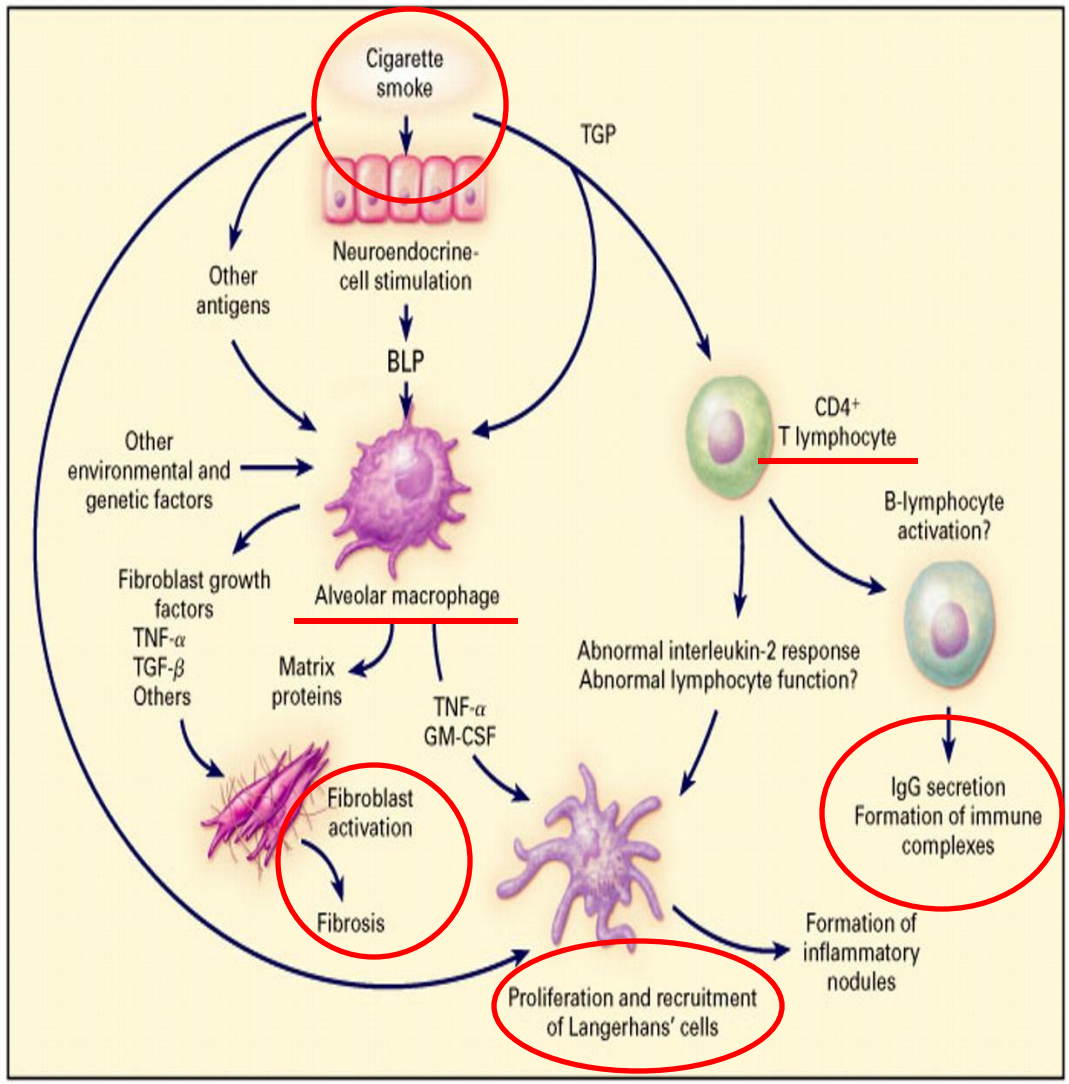
- **Pulmonary** LCH is almost exclusive to **smokers**
- Symptoms generally dissipate after **smoking cessation**

- Sometimes associated with a BRAF gene mutation
- Median age of diagnosis, **late30s-early40s**

Proposed Pathogenesis of Pulmonary Langerhans Cell Histiocytosis

Primary event: **Cigarette-smoke-induced recruitment** and activation of **Langerhans cells to lung**; potential mechanisms:

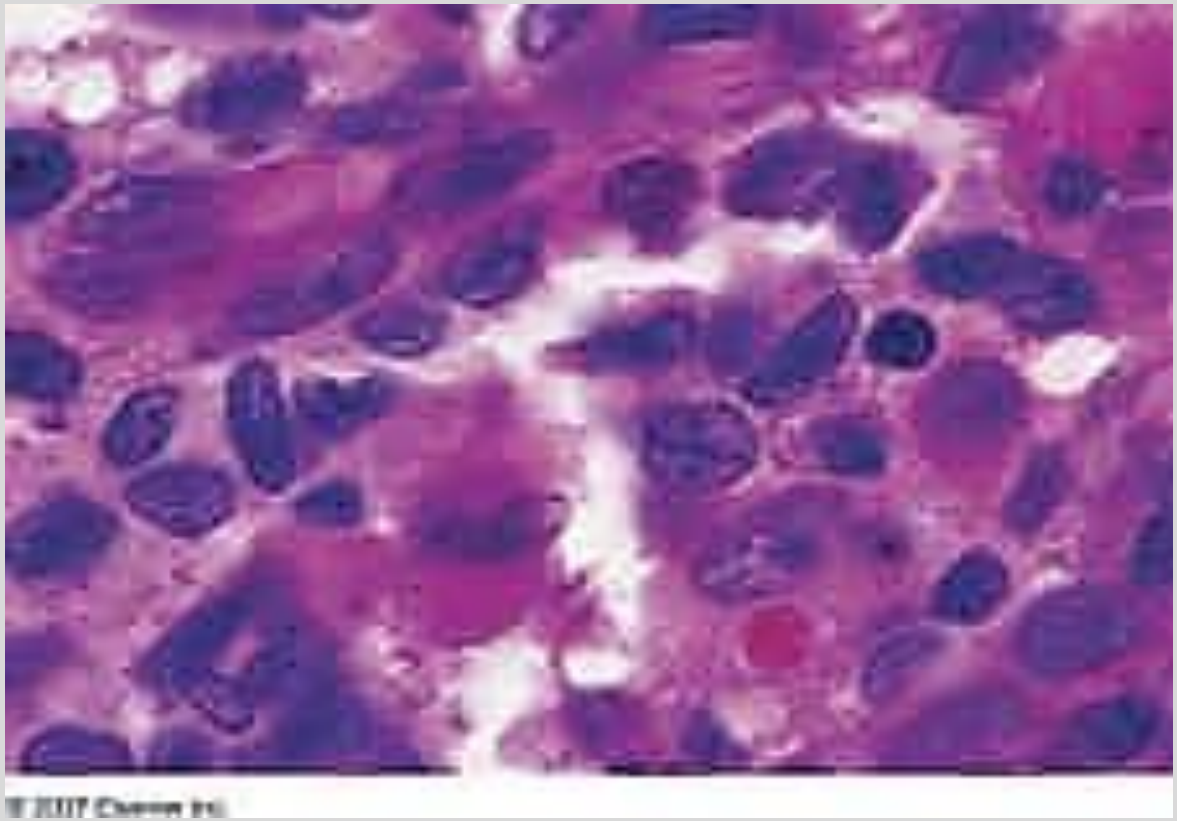
- Activate alveolar **macrophages**
- Macrophages produce cytokines → ↑ recruitment **Langerhans** cells activity
- Local expansion of **T** lymphocytes
- Stimulates **B**-lymphocyte → **antibodies** & immune-complex formation
- Fibroblast activation & **fibrosis**



Vassallo R et al. N Engl J Med 2000;342:1969-1978

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Prominent **longitudinal grooves** in the nuclei of Langerhans cells in a case of pulmonary Langerhans cell histiocytosis (H&E)

Electron microscopy (**EM**) **- Birbeck granules**

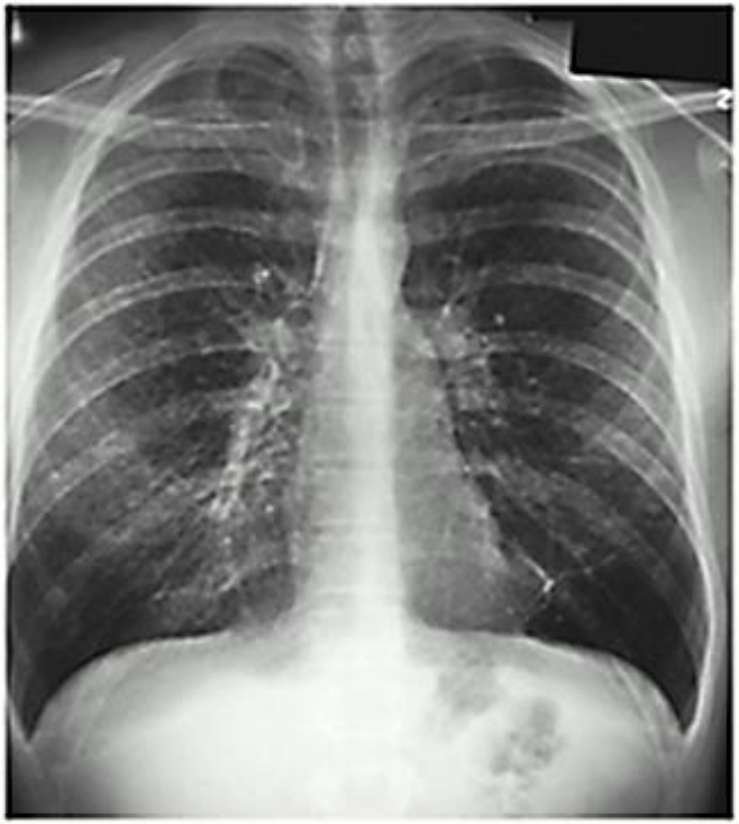


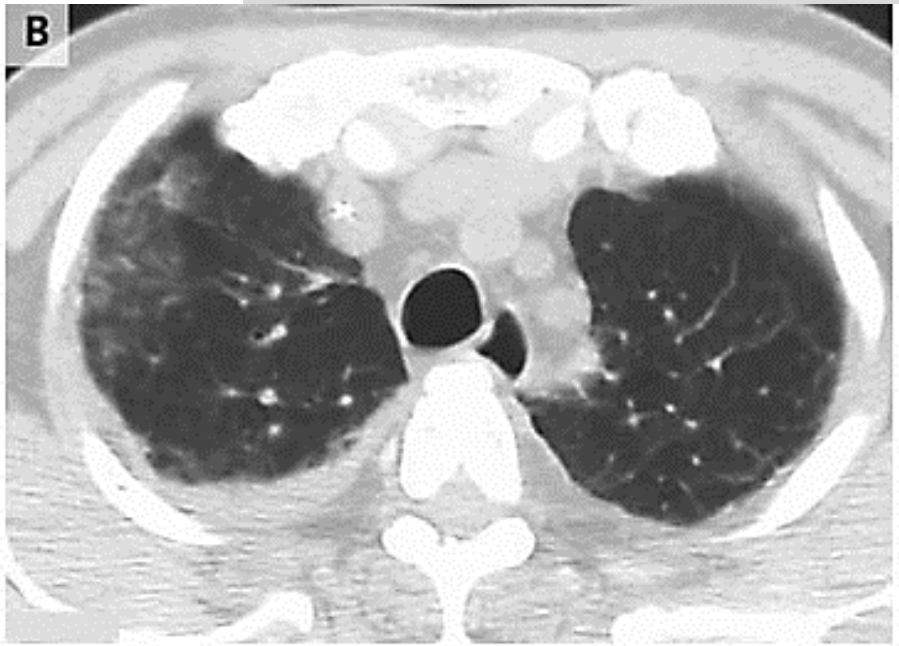
- Distinctive, **tennis racket-shaped** organelles found exclusively **within Langerhans cells**

http://www.humpath.com/spip.php?article5997

As lung lesions progress, **scarring** occurs causing airway destruction and alveolar damage

Chest X-ray of Pulmonary Langerhans Cell Histiocytosis w/ **Reticulonodular Infiltrate** Predominantly **upper- & middle**-lobes





CT image: **Ground-glass** opacity in right upper lobe & interlobular septal, & peri-bronchial thickening

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Hypersensitivity pneumonitis

- **A group** of interstitial lung disorders **- immune mediated,** caused by intense, usually long-term exposure to **inhaled organic dusts**
  - Organic dust usually contains antigens made of spores of **bacteria, bacterial products, fungi, animal proteins**
  - Most patients have specific serum **antibodies** against causative agent

- Symptoms typically 4-6 hours after exposure
  - May last ~ **12 hours-days** & **recur** w/ re-exposure

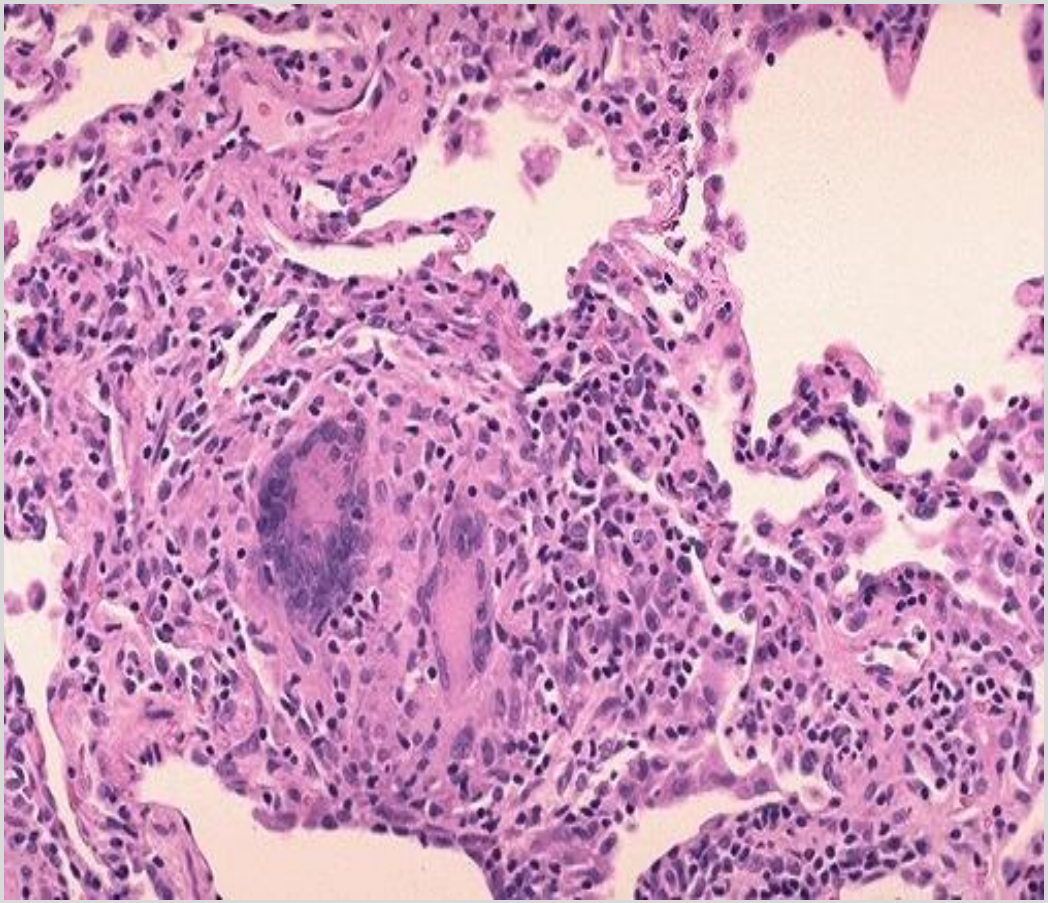
- Examples:
  - **Farmer's** lung (exposure to dusts from humid, warm, newly harvested **hay**, permits proliferation of spores of thermophilic (like warm temperatures) actinomycetes)
  - Pigeon breeder's/ Bird fancier's lung (provoked by proteins from serum, excreta, or feathers of **birds**)
  - Air conditioner/**hot tub** lung (caused by thermophilic **bacteria** in heated water reservoirs)

Hypersensitivity pneumonitis management & histopathology

- Serious **chronic fibrotic lung** disease may be prevented by **removing triggers**

- Histopathologic changes usually centered on bronchioles
  - **Noncaseating granulomas** in 2/3rds of patients, suggesting **T cell-mediated delayed-type** hypersensitivity reaction against antigen

Hypersensitivity pneumonitis histopathology



- H&E stain
- **Interstitial** pneumonitis w/ lymphocytes, plasma cells & macrophages (**noncaseating granulomas**)

- Interstitial fibrosis & obliterative bronchiolitis & honeycombing (**late** stage pathology) & fibroblastic foci (**new** pathologic injury)

http://library.med.utah.edu/WebPath/LUNGHTML/LUNG091.html

Pulmonary eosinophilic syndromes

- **Rare**, characterized by infiltration of **eosinophils**, recruited by elevated alveolar **IL-5**, & other eosinophil attractants:

  - **Acute eosinophilic pneumonia (AEP) with respiratory failure** – acute, **idiopathic**, rapid onset w/ fever, dyspnea, & respiratory failure
    - Histopathology shows diffuse alveolar damage & **many eosinophils** → responds to **corticosteroids**

  - **Chronic eosinophilic pneumonia (CEP)** – focal cellular consolidation, mostly **peripheral**, aggregates of **lymphocytes & eosinophils** within septal walls & alveoli, may have interstitial fibrosis
    - Signs & symptoms: Cough, fever, night sweats, dyspnea and weight loss → respond to **corticosteroids**
    - **Often idiopathic**

      ↓
  - **Secondary eosinophilia** – occurs in **parasitic**, **fungal**, and **bacterial** infections, hypersensitivity pneumonitis, **drug** allergies, & **eosinophilic granulomatosis w/ polyangiitis** (formerly known as allergic granulomatosis/Churg-Strauss syndrome - believed to be an autoimmune disorder in susceptible patients to an unknown trigger) → **manage underlying condition**

Pulmonary eosinophilic syndromes main diagnostic criteria:

- **Radiographically** identified pulmonary abnormalities need to be **combined** with other diagnostic methods:

- ↑ **peripheral** blood **eosinophilia** (>500 Eos/mm^3)

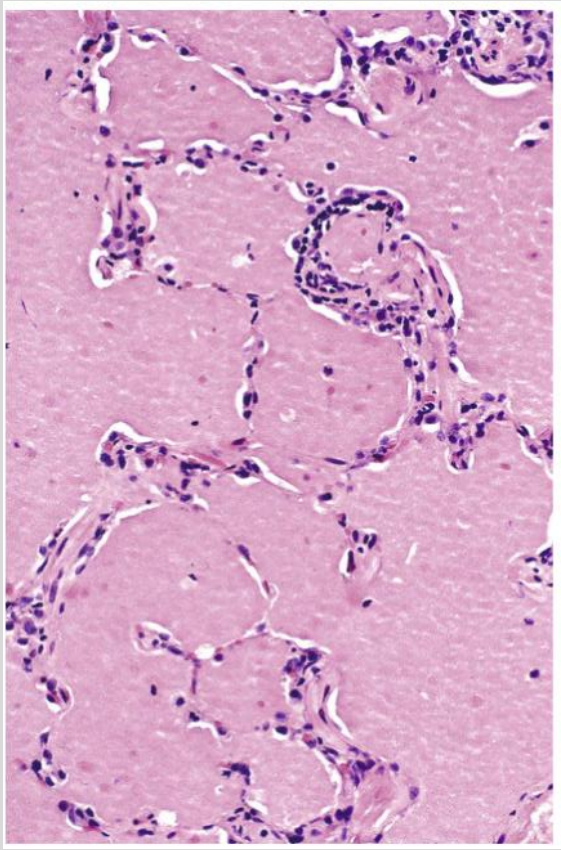
- ↑ eosinophils in **BALF**:
- cut-off of ≥25% for AEP diagnosis
- cut-off of ≥40% for CEP diagnosis

- Lung **tissue eosinophilia** demonstrated in transbronchial or open lung **biopsies**

- **Serology** can help to support diagnoses: CRP, IgE levels, ELISA for coccidioidomycosis, helminths infections for example

Pulmonary alveolar proteinosis (PAP)

- **Rare**
- Caused by deficient **GM-CSF/macrophage dysfunction**
  - **Results** in accumulation of **intra-alveolar precipitates** containing **surfactant proteins**
- **Lamellar bodies (EM)** -↑**normally** made **surfactant**-containing particles
  - Buildup of abnormal surfactant material rich in lamellar bodies that **fill alveoli**
- Precipitate is pink, homogeneous, **PAS+**
- Hereditary – **GM-CSF gene mutations** in neonates affect surfactant proteins
- **Autoimmune,** 90% of patients have antibodies against GM-CSF
- Treated with lung lavage, GM-CSF, secondary cause treatment



H&E stain, lung tissue

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